

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – IX

Roll No.: T005	Name: Ritesh Chaubey
Class: TYIT	Batch:1
Date of Assignment: 27/08/2026	Date/Time of Submission:27/08/2026

AIM: Cleaning and Structuring Data for Analysis.

A. Identify and handle missing or inconsistent data.

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. In the Connect pane on the left side, click Text File.
3. Select Sem1_Scores.csv and open it.
4. The dataset will appear in the Data Source tab.
5. Observe that:
 - Math contains missing (Null) values.
 - Physics contains percentage values (90%, 75%) and one missing value.
 - Chemistry also contains a missing value.
 - Physics and Attendance are shown as Text (Abc) because they contain %.

Sem1_Scores

Connection: ☒ Live ☐ Extract Filters: 0 | Add

Sem1_Scores.csv

Sem1_Scores.csv 8 fields 10 rows 10 rows

Name	Sem1_Scores.csv	Abc	Abc	Abc	#	#	#	#	Abc
Description		Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv	Sem1_Scores.csv
Fields		Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S101	Aarav Sharma	CSE-01		85	0.900000		88	95	85 %
S102	priya VERMA	ECE-02		null	0.750000		80	85	90 %
S103	Rohan Gupta	CSE-01		78	0.820000		null	88	78 %
S104	Sneha Patel	IT-03		92	0.880000		90	96	95 %
S105	aditya Rao	MECH-04		45	null		50	60	65 %
S106	Ananya Singh	CSE-01		88	0.910000		85	92	88 %
S107	Rahul Nair	ECE-02		null	0.700000		75	80	72 %
S108	Kavya Reddy	IT-03		95	0.940000		92	98	98 %
S109	Vikram Joshi	CSE-01		67	0.600000		58	72	80 %
S110	Pooja Iyer	ECE-02		82	0.850000		80	89	91 %

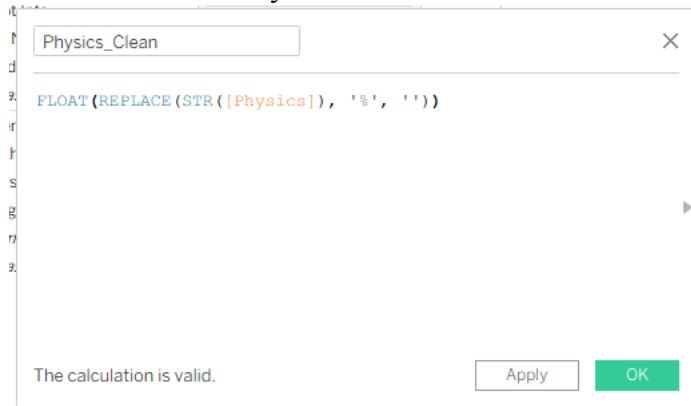
Step 2: Create Physics_Clean (Convert Physics to Numeric)

Since the Physics column contains %, it is treated as text. Create a calculated field to convert it into a numeric value.

Procedure

1. Click Sheet 1.
2. In the Data Pane, right-click on an empty area.
3. Select Create → Calculated Field.
4. Enter the field name: **Physics_Clean**
5. Enter the following formula: `FLOAT(REPLACE(STR([Physics]), '%', ''))`
6. Click OK.

This removes the % symbol and converts the value into a number.



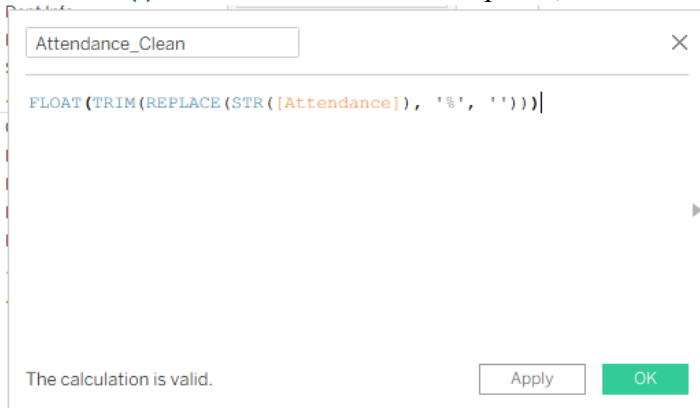
Step 3: Create Attendance_Clean

Attendance also contains %, so convert it into numeric values.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name the field: **Attendance_Clean**
4. Enter the formula: `Float (TRIM (REPLACE (STR ([Attendance]), '%', '')))`
5. Click OK.

The **TRIM()** function removes extra spaces, while **REPLACE()** removes %.



Step 4: Handle Missing Values in Math

Replace Null values in Math with 0 using **IFNULL()**.

Procedure

1. Right-click the Data Pane.
2. Choose Create → Calculated Field.
3. Name the field: **Math_Clean**
4. Enter the formula: `IFNULL([Math], 0)`
5. Click OK.

This replaces every missing Math value with 0



Step 5: Handle Missing Values in Chemistry

Replace missing Chemistry values with 0.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name it: **Chemistry_Clean**
4. Enter: IFNULL([Chemistry], 0)
5. Click OK.

This fills all missing Chemistry values with 0.



Step 6: Update Physics_Clean to Handle Null Values

Student S105 has a missing Physics value. Update the earlier calculated field to replace Null with 0.

Procedure

1. In the Data Pane, right-click Physics_Clean.
2. Select Edit.
3. Replace the previous formula with: IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)
4. Click OK.

Now the field performs two tasks:

- Removes %
- Converts text to numeric
- Replaces Null values with 0.



Part A Insight:

1. The **Sem1_Scores.csv** dataset contained missing (Null) values in the Math, Physics, and Chemistry columns, which could affect calculations and visualizations.
2. The Physics and Attendance columns were stored as Text (Abc) because they contained % symbols, preventing Tableau from treating them as numerical measures.
3. Using **REPLACE()**, **TRIM()**, and **FLOAT()**, the percentage values were converted into numeric values, making them suitable for calculations like averages and charts.
4. The **IFNULL()** function replaced missing values with 0, ensuring that no records were ignored during analysis.
5. After cleaning, all required fields became consistent and ready for accurate visualization and further data analysis in Tableau.

B. Convert wide data into tall format.

Step 1: Open the Data Source Grid

1. Open the Sem1_Scores.csv dataset in Tableau.
2. Click the Data Source tab at the bottom-left corner.
3. Locate the four subject columns:
 - Math
 - Physics
 - Chemistry
 - Programming

Name Sem1_Scores.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remote Fiel...
Abc	Dept Info ▼	Sem1_Scores.csv	Dept_Info
#	Math ▼	Sem1_Scores.csv	Math
#	Physics ▼	Sem1_Scores.csv	Physics
#	Chemistry ▼	Sem1_Scores.csv	Chemistry
#	Programming ▼	Sem1_Scores.csv	Programming
Abc	Attendance ▼	Sem1_Scores.csv	Attendance
=#	Physics_Clean ▼	Calculation	Calculation_0...
=#	Attendance_C ▼	Calculation	Calculation_0...
=#	Math_Clean ▼	Calculation	Calculation_0...

Step 2: Multi-Select and Pivot the Columns

1. Click the Math column header.
2. Hold down the Ctrl key.
3. Select Physics, Chemistry, and Programming.
4. Ensure all four columns are highlighted in blue.
5. Right-click any highlighted column.
6. Select Pivot from the menu.

After clicking Pivot, Tableau combines the four columns into two new columns.

Student ID	Full Name	Dept Info	Attendance	Pivot Field Names	Pivot Field Values	Physics_Clean	Attendar
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	
S102	priya VERMA	ECE-02	90 %	Math	null	null	
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null	
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null	
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Physics	0.8200	null	
S103	Rohan Gupta	CSE-01	78 %	Programming	88.0000	null	
S104	Sneha Patel	IT-03	95 %	Chemistry	90.0000	null	

Step 3: Rename the Pivoted Fields

1. Double-click Pivot Field Names.
2. Rename it to Subject.
3. Double-click Pivot Field Values.
4. Rename it to Score.

Now the dataset becomes easier to analyze because all subjects are stored in a single column

Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Clean	Attendar
S101	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	
S101	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	
S101	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	
S102	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	
S102	priya VERMA	ECE-02	90 %	Math	null	null	
S102	priya VERMA	ECE-02	90 %	Physics	0.7500	null	
S102	priya VERMA	ECE-02	90 %	Programming	85.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Chemistry	null	null	
S103	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	
S103	Rohan Gupta	CSE-01	78 %	Physics	0.8200	null	
S103	Rohan Gupta	CSE-01	78 %	Programming	88.0000	null	
S104	Sneha Patel	IT-03	95 %	Chemistry	90.0000	null	

Step 4: Convert Score into a Numeric Measure

After pivoting, the Score field may still appear as Text (Abc) because Physics originally contained % symbols. Convert it into a numeric field.

Method 1 (Direct Conversion)

1. Click the Abc icon above the Score column.
2. Select Number (Decimal)

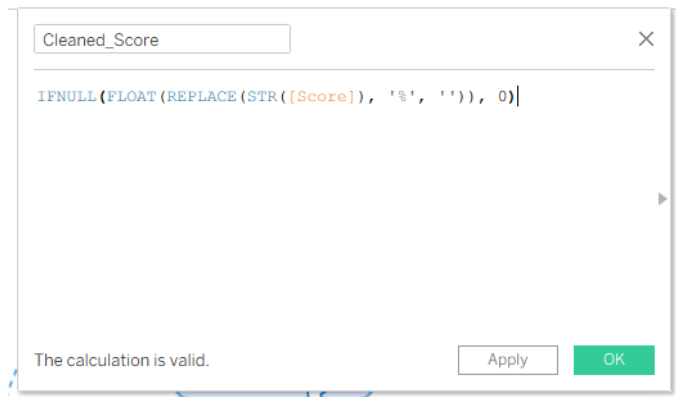
Method 2 (Calculated Field)

If the Score field still behaves as text:

1. Open Sheet 1.
2. Right-click the Data Pane.
3. Select Create → Calculated Field.
4. Enter:
 - Field Name: **Clean_Score**
5. Use this formula: `IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)`
6. Click OK.

This formula:

- Removes %
- Converts text to numbers
- Replaces Null values with



Insight (Part B)

1. The original dataset stored Math, Physics, Chemistry, and Programming as separate columns, making subject-wise comparison less flexible.
2. Using the Pivot feature converted the dataset from Wide Format to Tall Format, creating two new fields: Subject and Score.
3. Converting the Score field into a numeric measure removed text-related limitations caused by % symbols.
4. The cleaned tall-format dataset is better suited for creating dynamic visualizations such as subject-wise bar charts, filters, and comparisons in Tableau.

C. Perform union of multiple files

Step 1: Open the Data Source Canvas

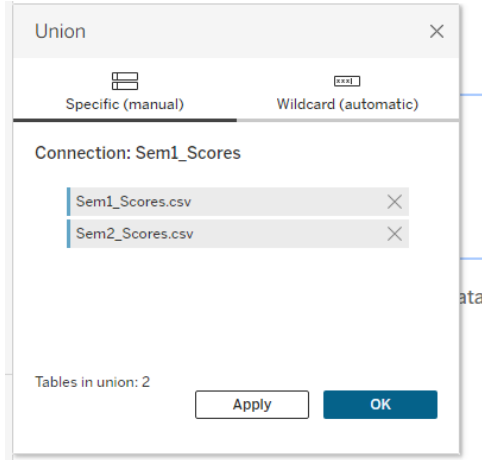
1. Open Tableau Desktop.
2. Click the Data Source tab at the bottom-left.
3. If a table is already present on the canvas, right-click it and select Remove.
4. Keep the canvas empty before creating the Union

Step 2: Create a Multi-File Union

1. In the left sidebar, locate Files / Connections.
2. Drag New Union onto the empty canvas.
3. The Union dialog box will open.

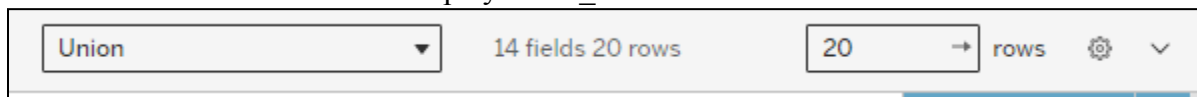
4. Drag Sem1_Scores.csv into the Union box.
5. Drag Sem2_Scores.csv into the same Union box.
6. Verify that both file names appear in the dialog.
7. Click Apply.
8. Click OK.

After applying the Union, Tableau creates one combined table represented by a plus (+) symbol.



Step 3: Verify the Merged Data

1. Look at the data grid at the bottom.
2. Confirm that the total number of rows has increased from 10 to 20.
3. Scroll to the far-right side.
4. Verify the Table Name column.
5. Check that:
 - First 10 records display Sem1_Scores.csv
 - Next 10 records display Sem2_Scores.csv



Step 4: Create a Semester Identifier

Create a calculated field to identify whether each record belongs to Semester 1 or Semester 2.

Procedure

1. Click Sheet 1.
2. In the Data Pane, right-click an empty space.
3. Select Create → Calculated Field.
4. Enter the field name: **Semester**
5. Enter the formula:

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

6. Click OK.

This automatically labels every record according to its original file.



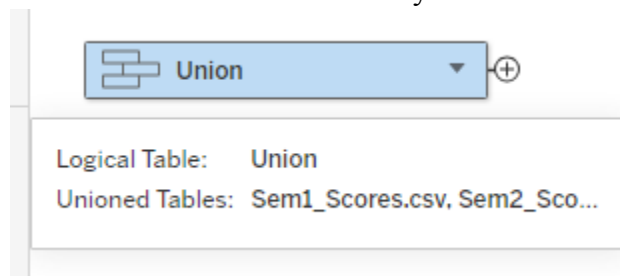
Insight (Part C)

1. The Union feature combined Sem1_Scores.csv and Sem2_Scores.csv into a single dataset without changing their structure.
2. The total number of student records increased from 10 to 20, allowing both semesters to be analyzed together.
3. The automatically generated Table Name field preserved the source of each record, making it easy to distinguish between Semester 1 and Semester 2.
4. Creating the Semester calculated field made it easier to compare student performance across different semesters using a single dataset.

D. Execute cross-database joins

Step 1: Keep the Union Table on the Canvas

1. Open the workbook created in Part C.
2. Go to the Data Source tab.
3. Ensure the Union table (Sem1 + Sem2) is already present on the canvas.
4. Do not remove or modify the Union table before creating relationships.



Step 2: Add and Link Department_Master.csv

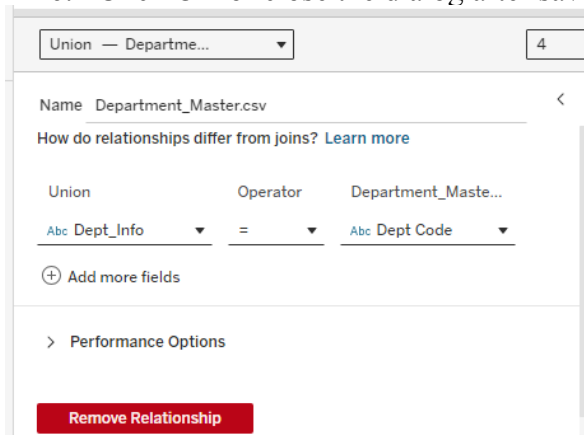
1. In the Files / Connections pane, locate Department_Master.csv.
2. Drag it to the right side of the Union table.
3. Tableau creates a relationship line between the two tables.
4. Click the relationship line if the dialog does not open automatically.
5. Set the relationship as follows:

Field	Value
Left Table	Union
Left Field	Dept_Info
Right Table	Department_Master

Right Field **Dept_Code**

Operator **=**

6. Click OK or close the dialog after saving the relationship.



Step 3: Add and Link Student_Activities.csv

1. In the Files / Connections pane, locate Student_Activities.csv.
2. Drag it onto the canvas.
3. Drop it beside the Union table.
4. Tableau opens the relationship dialog.
5. Configure the relationship:

Field Value

Left Table Union

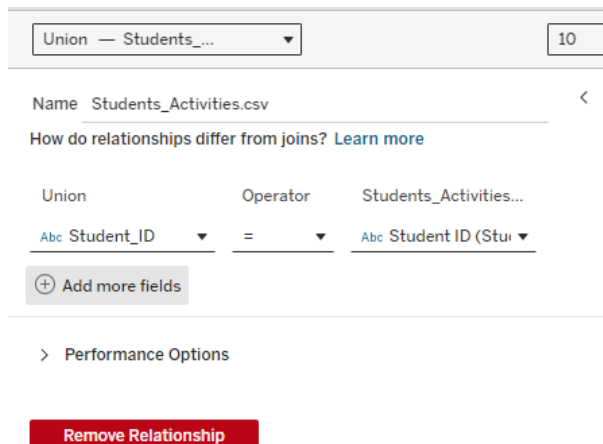
Left Field **Student_ID**

Right Table Student_Activities

Right Field **Student_ID**

Operator **=**

6. Click OK.



Step 4: Verify the Joined Data in Sheet 1

1. Click Sheet 1.
2. Look at the Data Pane on the left.
3. Verify that fields from all three tables are available.

Union Table Fields

- Student_ID
- Full_Name
- Dept_Info
- Math
- Physics
- Chemistry
- Programming
- Attendance
- Table Name

Department Master Fields

- Dept_Code
- Department_Name
- HOD_Name
- Dept_Target_Average
- Total_Dept_Capacity

Student Activities Fields

- Club_Membership
- Projects_Completed
- Certification

The screenshot shows a data visualization tool interface. On the left is the 'Tables' pane, which lists three data sources: 'Department_Master.csv', 'Students_Activities.csv', and 'Union'. Each source has a list of fields. The 'Union' table is currently selected, and its fields are: 'Attendance', 'Dept_Info', 'Full_Name', 'Semester' (highlighted with a dropdown arrow), 'Student_ID', 'Table Name', 'Attendance_Clean', 'Chemistry', 'Chemistry_Clean', 'Math', 'Math_Clean', 'Physics', and 'Physics_Clean'. On the right is the 'Marks' card, which has a dropdown menu set to 'Automatic' and several icons for 'Color', 'Size', 'Text', 'Detail', and 'Tooltip'. The 'Filters' pane is empty. The 'Sh' (Sheet) pane is partially visible on the far right.

Insight (Part D)

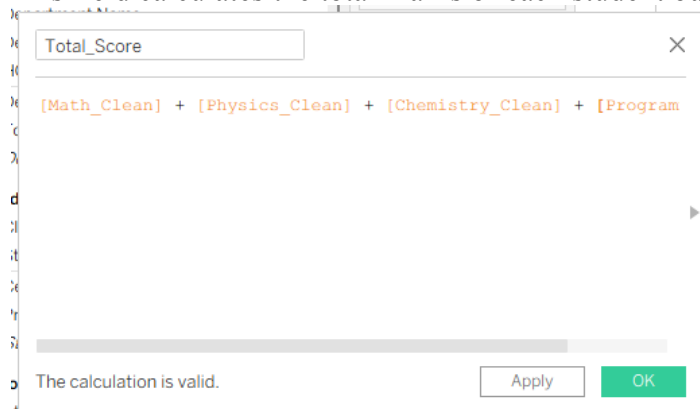
1. Cross-database relationships connected the Unioned Student Scores dataset with two independent tables using common keys.
2. The relationship **Dept_Info = Dept_Code** added department details such as Department Name, HOD Name, and Department Target Average to the student records.
3. The relationship **Student_ID = Student_ID** added extracurricular information like Club Membership, Projects Completed, and Certifications.
4. After the relationships were created, all three datasets could be analyzed together, making it easier to build richer dashboards combining academic, departmental, and activity data.

E. Restructure data for better visualization

Step 1: Create Total Score

1. Open Sheet 1 in Tableau.
2. In the Data Pane, right-click an empty space.
3. Select Create → Calculated Field.
4. Enter the field name: **Total_Score**
5. Enter the formula: $[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]$
6. Click OK.

This field calculates the total marks of each student out of 400.



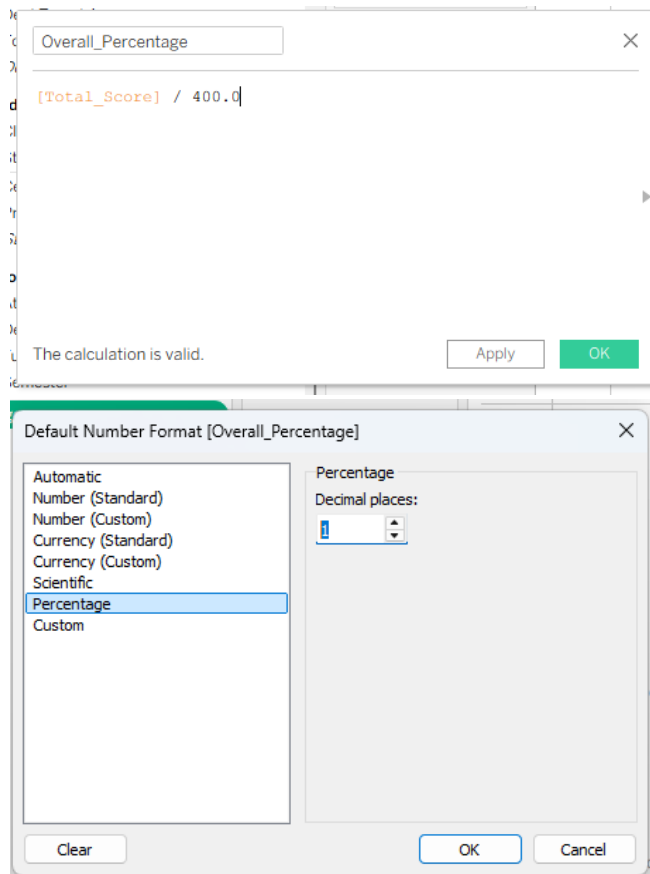
Step 2: Create Overall Percentage

1. Right-click the Data Pane again.
2. Select Create → Calculated Field.
3. Enter the field name: **Overall_Percentage**
4. Enter the formula: $[Total_Score] / 400.0$
5. Click OK.

This converts the total score into an overall percentage.

Format the Percentage

1. Right-click Overall_Percentage.
2. Select Default Properties → Number Format.
3. Choose Percentage.
4. Set Decimal Places = 1.
5. Click OK.



Step 3: Create Performance Band

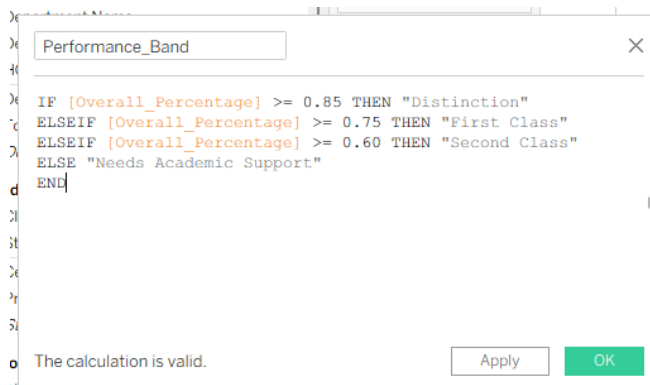
Create a grading category for every student based on their percentage.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Performance_Band**
4. Enter the formula:

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"  
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"  
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"  
ELSE "Needs Academic Support"  
END
```

5. Click OK.



Step 4: Create Attendance Status

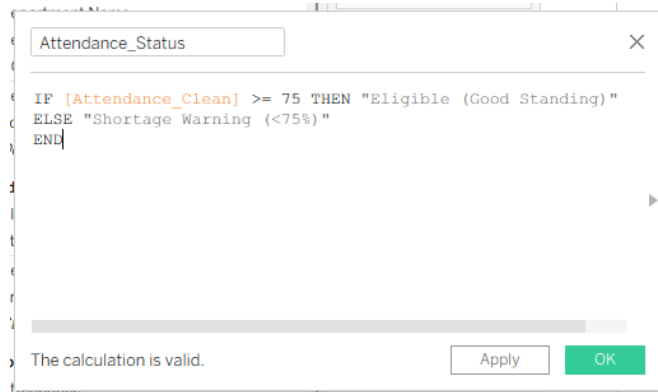
Create an attendance indicator to identify whether a student has sufficient attendance.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Attendance_Status**

4. Enter the formula:
 IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
 ELSE "Shortage Warning (<75%)"
 END

5. Click OK.



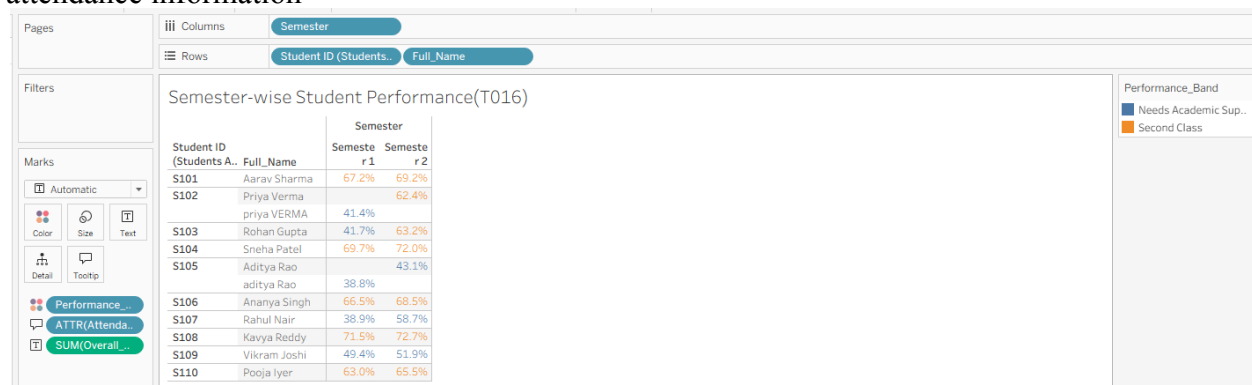
Step 5: Build the Student Performance Scorecard

Create a simple dashboard-like scorecard using the calculated fields.

Procedure

1. Drag Student_ID to the Rows shelf.
2. Drag Full_Name to the Rows shelf.
3. Drag Semester to the Columns shelf.
4. Drag Overall_Percentage to the Text option on the Marks card.
5. Drag Performance_Band to the Color option.
6. Drag Attendance_Status to the Tooltip (or Detail) option.

The scorecard will display each student's percentage with color-coded performance and attendance information



Insight (Part E)

1. The **Total_Score** field combines marks from all four subjects into a single value, making overall performance easier to measure.
2. **Overall_Percentage** converts total marks into a standardized percentage, allowing fair comparison between students.

3. **Performance_Band** automatically classifies students into Distinction, First Class, Second Class, or Needs Academic Support, making performance easier to interpret.
4. **Attendance_Status** highlights whether students meet the 75% attendance requirement, helping identify students who need attention.
5. The final scorecard combines academic performance and attendance into one view, making the dashboard more informative and easier to analyze

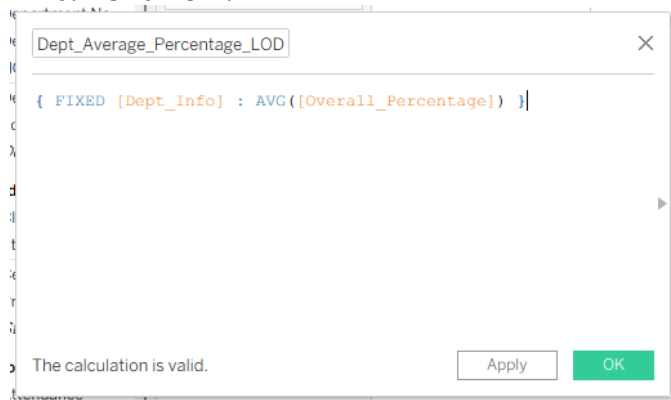
F. Handle different levels of detail in datasets

Step 1: Create Department-Level Benchmark (LOD)

This calculates the average percentage for each department using a FIXED LOD expression.

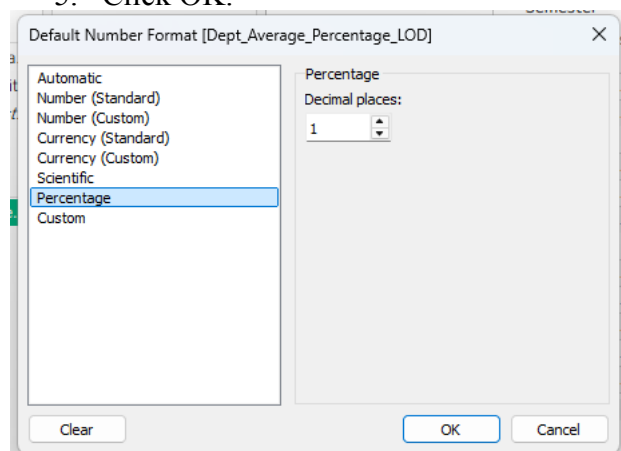
Procedure

1. Open Sheet 1 (or create a new worksheet).
2. In the Data Pane, right-click an empty space.
3. Select Create → Calculated Field.
4. Enter the field name: **Dept_Average_Percentage_LOD**
5. Enter the formula: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
6. Click OK.



Format the Field

1. Right-click **Dept_Average_Percentage_LOD**.
2. Select Default Properties → Number Format.
3. Choose Percentage.
4. Set Decimal Places = 1.
5. Click OK.

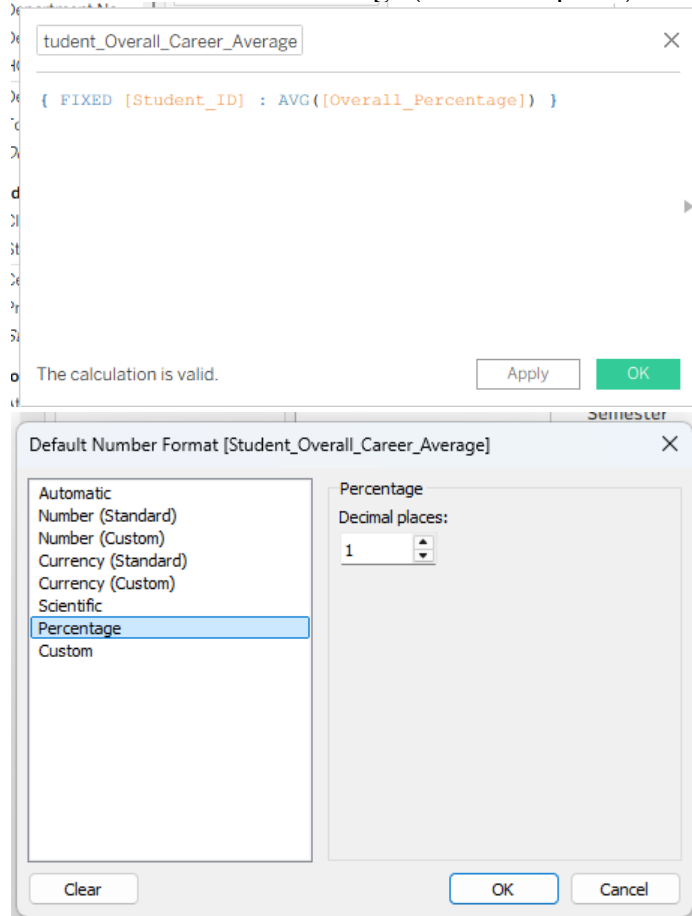


Step 2: Create Student Career Average (LOD)

This calculates each student's average percentage across all semesters.

Procedure

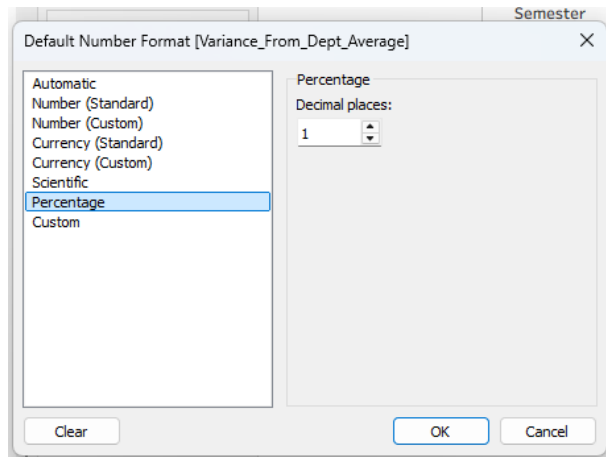
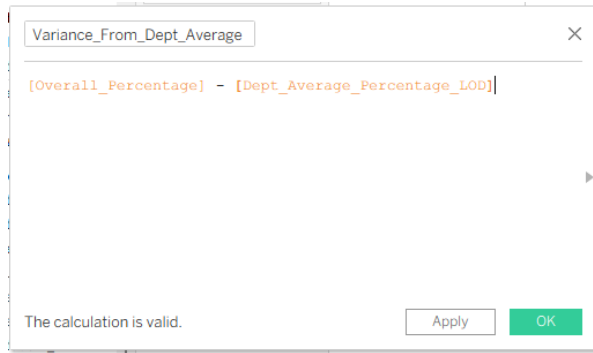
1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Student_Overall_Career_Average**
4. Enter the formula: { FIXED [Student_ID] : AVG([Overall_Percentage]) }
5. Click OK.
6. Format it as Percentage (1 decimal place).

**Step 3: Create Performance Variance**

Calculate how much each student's percentage differs from their department average.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Variance_From_Dept_Average**
4. Enter the formula: [Overall_Percentage] - [Dept_Average_Percentage_LOD]
5. Click OK.
6. Format it as Percentage (1 decimal place).



Step 4: Create Benchmark Status

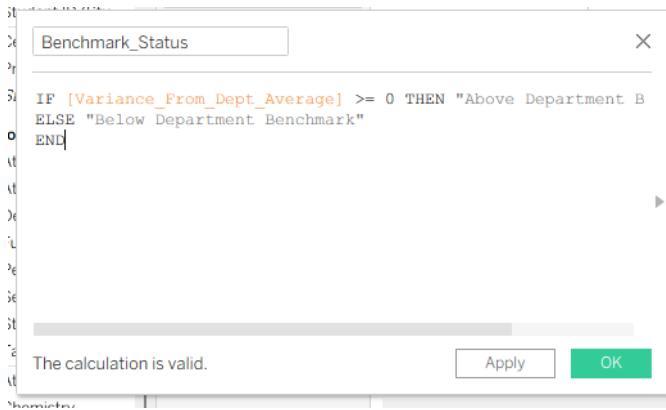
Create a label that identifies whether a student is above or below the department benchmark.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Benchmark_Status**
4. Enter the formula:

```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"  
ELSE "Below Department Benchmark"  
END
```

5. Click OK



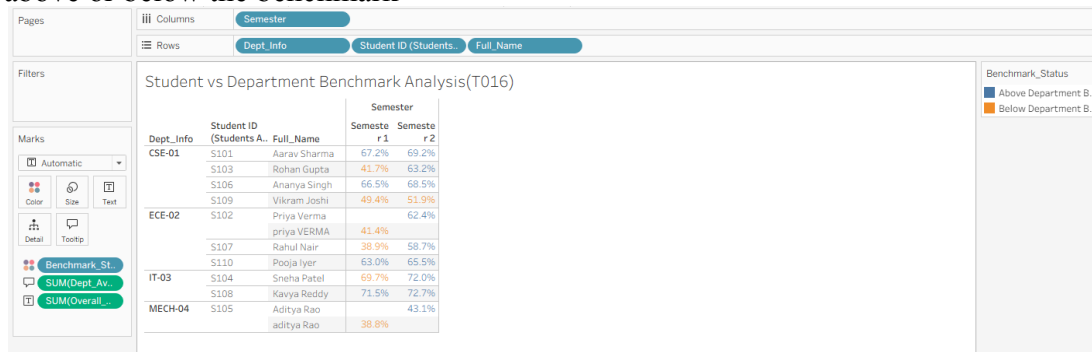
Step 5: Build the LOD Benchmark Visualization

Create a visualization that compares each student's performance with the department benchmark.

Procedure

1. Drag Dept_Info to the Rows shelf.
2. Drag Student_ID to the Rows shelf.
3. Drag Full_Name to the Rows shelf.
4. Drag Semester to the Columns shelf.
5. Drag Overall_Percentage to the Text option on the Marks card.
6. Drag Dept_Average_Percentage_LOD to Tooltip.
7. Drag Benchmark_Status to Color.

The visualization will show each student's percentage, department average, and whether they are above or below the benchmark



Insight (Part F)

1. FIXED LOD expressions calculate department averages without changing individual student records.
2. **Student_Overall_Career_Average** combines each student's performance across multiple semesters into a single benchmark.
3. **Variance_From_Dept_Average** shows how much a student's performance differs from the department average.
4. **Benchmark_Status** makes it easy to identify students who are performing above or below their department benchmark.
5. The final LOD visualization compares individual performance with department-level performance in a single view, making analysis more meaningful

G. Apply advanced data cleaning techniques

Step 1: Clean Student Names

The **Full_Name** field contains extra spaces and inconsistent letter casing (for example, " Aarav Sharma" and "priya VERMA"). Create a cleaned version of the name.

Procedure

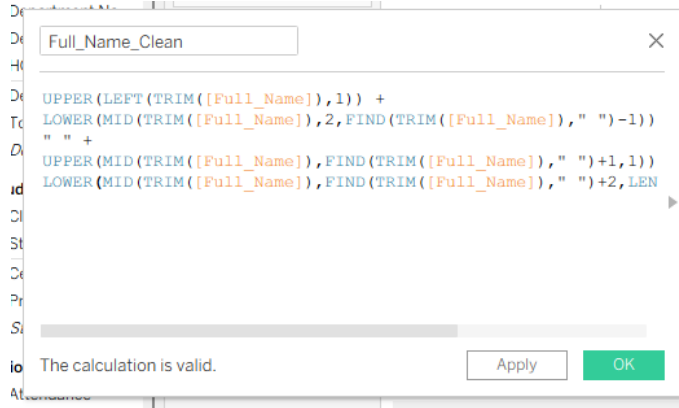
1. Open Sheet 1.
2. In the Data Pane, right-click an empty space.
3. Select Create → Calculated Field.
4. Enter the field name: **Full_Name_Clean**
5. Enter the formula:

```
UPPER(LEFT(TRIM([Full_Name]),1)) +
LOWER(MID(TRIM([Full_Name]),2,FIND(TRIM([Full_Name])," ")-1)) +
" " +
UPPER(MID(TRIM([Full_Name]),FIND(TRIM([Full_Name])," ")+1,1)) +
LOWER(MID(TRIM([Full_Name]),FIND(TRIM([Full_Name]),"
")+2,LEN(TRIM([Full_Name]))))
```

Alternative (Simple Method)

TRIM([Full_Name])

6. Click OK.

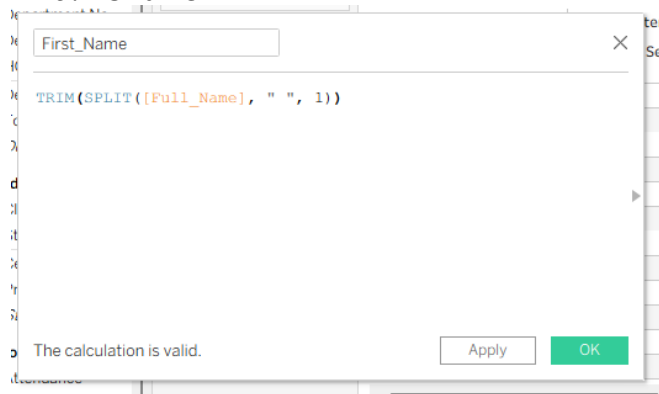


Step 2: Extract First Name and Last Name

Use the SPLIT() function to separate the student's first and last names.

Step 2(A): Create First Name

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **First_Name**
4. Enter the formula: TRIM(SPLIT([Full_Name], " ", 1))
5. Click OK



Step 2(B): Create Last Name

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Last_Name**
4. Enter the formula: TRIM(SPLIT([Full_Name], " ", -1))
5. Click OK

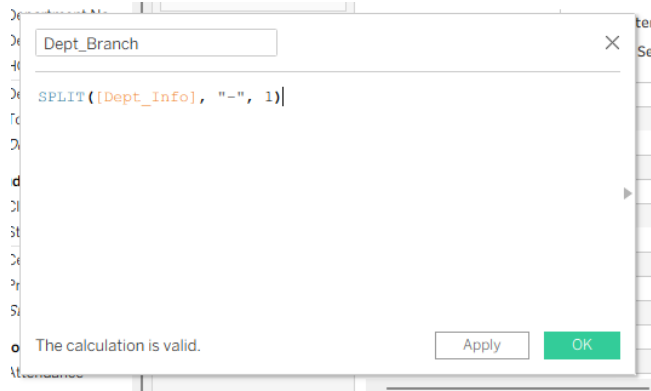


Step 3: Split Department Code

The **Dept_Info** field contains combined values like CSE-01, ECE-02, and MECH-04. Split them into Department Branch and Section ID.

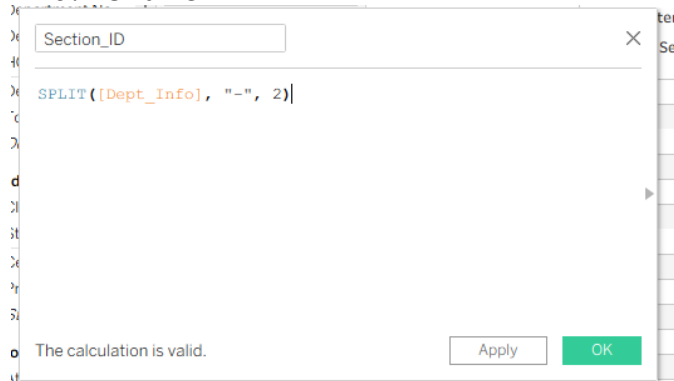
Step 3(A): Create Department Branch

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Dept_Branch**
4. Enter the formula: `SPLIT([Dept_Info], "-", 1)`
5. Click OK.



Step 3(B): Create Section ID

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Section_ID**
4. Enter the formula: `SPLIT([Dept_Info], "-", 2)`
5. Click OK



Step 4: Create Student Display Label

Create a combined label that displays the student ID, cleaned name, and department branch together.

Procedure

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Enter the field name: **Student_Display_Label**
4. Enter the formula: `[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"`
5. Click OK



Step 5: Build the Final Dashboard View

Create a clean dashboard-style view using the cleaned fields.

Procedure

1. Drag Student_Display_Label to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage to Text on the Marks card.
4. Drag Performance_Band to Color.
5. Drag Dept_Branch and Section_ID to the Filters shelf.
6. Select any department or section from the filter to view the required students.

